

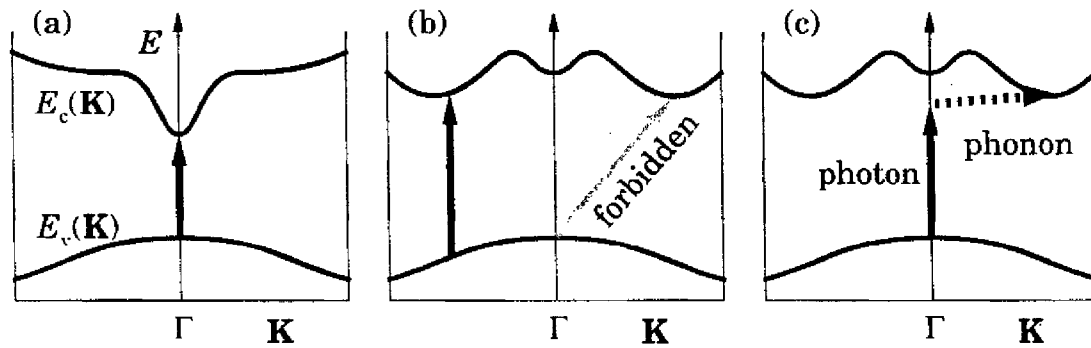


FIGURE 2.20. Optical absorption across the band gap in different types of semiconductor. (a) Absorption across a direct band gap at Γ . (b) Absorption across an indirect band gap is forbidden but vertical transitions occur for all K . (c) Transition across an indirect band gap with absorption of both a phonon and a photon.

of light. Most optoelectronic devices, and lasers in particular, are therefore based on materials such as GaAs with a direct gap.

In a material with an indirect gap such as AlAs, Si, or Ge, the extrema of the valence and conduction bands do not occur at the same K (Figure 2.20(b)). A transition between these points requires a large change in momentum as well as energy and cannot occur with a photon alone. Vertical transitions can occur at all points in the Brillouin zone, but the lowest energy of these optical transitions is greater than the minimum band gap.

Optical transitions across indirect gaps can occur if another process provides the change in momentum. Phonons can do this and either carry away or contribute a small amount of energy to the transition, as shown in Figure 2.20(c). Impurities may also be involved. These processes produce a tail in the optical absorption which starts around the energy of the indirect gap. Radiative recombination can occur by the same routes, but is much less efficient than that in materials with a direct gap. Thus Si, Ge, and their alloys are not generally efficient emitters of light. However, some light-emitting diodes (LEDs) use GaP, which has an indirect gap, and employ radiative recombination mediated by impurities to generate light.

2.8 Phonons

Phonons are the quanta of vibrations of the ions of the lattice away from their positions of minimum energy. The periodicity of the lattice means that phonons have band structure in much the same way as electrons. In some ways the band structure of phonons is more complicated than that for electrons, because phonons are polarized. Consider vibrations of very long wavelength, much greater than the lattice constant. These can be treated as classical sound waves, and there are three of them that propagate in a given direction. If we choose a simple direction, such as $[100]$ in a cubic material, one mode is longitudinal and the other two are transverse.

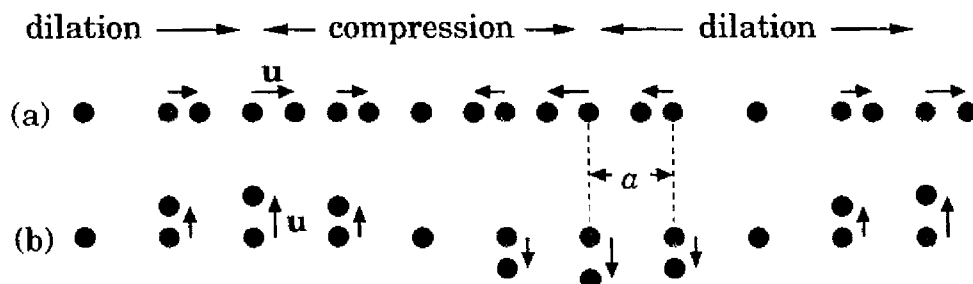


FIGURE 2.21. (a) Longitudinal and (b) transverse modes of the same wavelength $8a$ in a one-dimensional monatomic crystal. The light grey discs show the positions of the atoms at equilibrium, and the dark discs are their positions when displaced through $\mathbf{u}$.

These are illustrated for a one-dimensional crystal in Figure 2.21. The longitudinal mode is like a sound wave in air: the ions vibrate back and forth in the same direction $\mathbf{q}$ as the wave travels, so the displacement $\mathbf{u}$ is parallel to $\mathbf{q}$. This gives alternating zones of compression and dilation, in quadrature with the displacement. The ions vibrate in the plane normal to the direction of travel in the transverse modes, with $\mathbf{u}$ perpendicular to $\mathbf{q}$, like an electromagnetic wave in free space.

In other respects phonons are simpler than electrons. For example, the number of bands is given by the number of atoms per unit cell, multiplied by three for the polarizations, unlike the infinite number of bands for electrons. We shall solve a simple one-dimensional model that displays most of the main features of phonons, before looking briefly at phonons in a semiconductor.

2.8.1 PHONONS IN ONE DIMENSION

The simplest model is a one-dimensional row of atoms of mass m along x , separated by a distance a at rest, shown in Figure 2.22. They are connected by springs of elastic constant K . Vibration is restricted to the x -direction so there are only longitudinal modes. Let the displacement of atom j be u_j . Newton's law gives

$$m \frac{d^2 u_j}{dt^2} = K[(u_{j-1} - u_j) - (u_j - u_{j+1})], \quad (2.24)$$

where the terms in parentheses are the decreases in length of the bonds between atoms j and $j \pm 1$. We expect the solutions to be waves and can try a solution $u_j = U_0 \cos(qx - \omega_q t) = U_0 \operatorname{Re}\{\exp[i(qx - \omega_q t)]\}$, where $x = ja$ for atom j . The complex notation simplifies manipulation but is not essential as in the Schrödinger

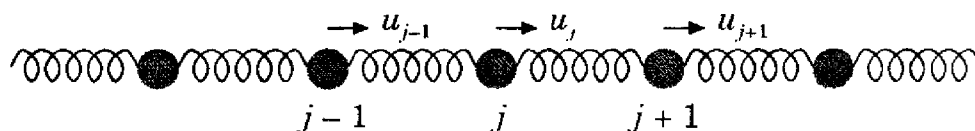


FIGURE 2.22. A one-dimensional row of springs and masses, labelled by j , allowed to vibrate longitudinally through displacements u_j .

equation. Substituting it into the equation of motion gives

$$-m\omega_q^2 U_0 e^{i(qja - \omega_q t)} = U_0 K [(e^{iq(j-1)a} - e^{iqja}) - (e^{iqja} - e^{iq(j+1)a})] e^{-i\omega_q t}. \quad (2.25)$$

The dependence on both position j and time t cancels to show that this solution is consistent, leaving

$$m\omega_q^2 = -K[(e^{-iqa} - 1) - (1 - e^{iqa})] = 4K \sin^2 \frac{qa}{2}. \quad (2.26)$$

We could have reached this more quickly by using Bloch's theorem in the form $u_{j+1} = e^{iqa} u_j$. The dispersion relation is

$$\omega_q = 2\sqrt{\frac{K}{m}} \left| \sin \frac{qa}{2} \right|. \quad (2.27)$$

The wave number should be restricted to the first Brillouin zone $|q| < \pi/a$, as with electrons. The dispersion is plotted in Figure 2.23(a). The relation is linear for small q , in contrast to the parabolic relation for electrons, and can be written $\omega_q = v_s q$, where the velocity of sound $v_s = a\sqrt{K/m}$. Adjacent atoms have similar displacements in this region of small q and long wavelength, as in Figure 2.21(a). The slope of ω_q decreases as q increases and becomes flat at the zone boundary $q = \pm\pi/a$, where the group velocity of the waves drops to zero as for electrons.

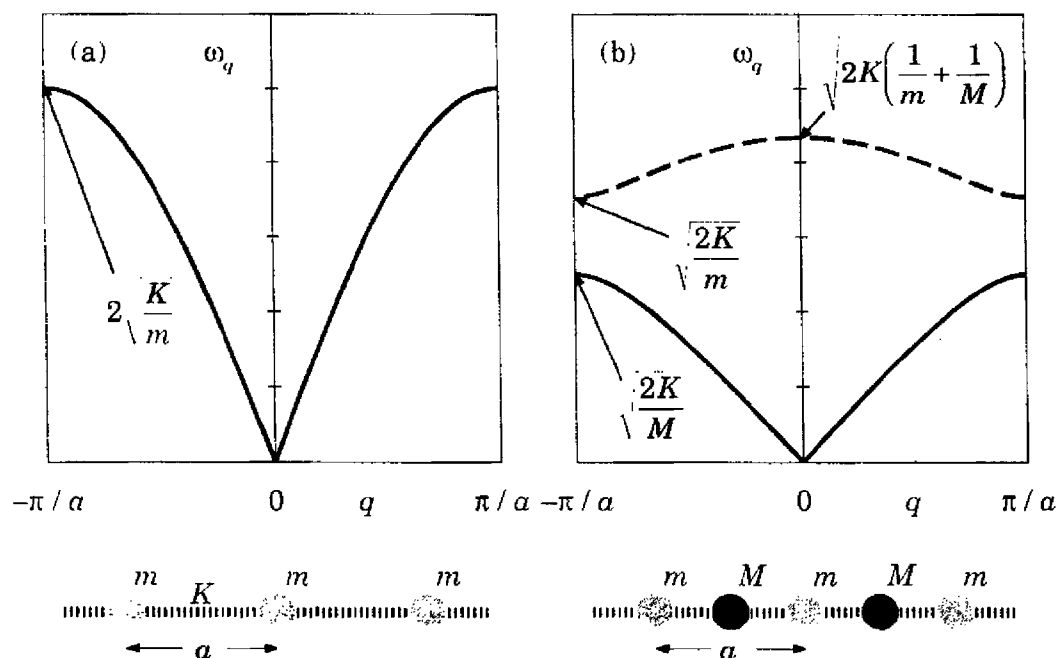


FIGURE 2.23. Dispersion relation ω_q for a one-dimensional crystal with longitudinal vibrations. (a) Monatomic crystal with atoms of mass m . (b) Diatomic chain of masses m and $M = 2m$, showing the acoustic branch (full line) and the optic branch (broken line). The scales are the same for both plots. The atomic configurations are shown underneath; both chains have the same lattice constant a and all springs have the same elastic constant K .

Also, atoms in adjacent cells vibrate in antiphase at this point. Alternate springs are fully compressed and stretched, giving the maximum frequency of vibration.

2.8.2 AMPLITUDE OF VIBRATION

We need a value for the amplitude of vibration U_0 to complete the analysis. Unfortunately we have hit a fundamental problem. The equations of motion are linear in the displacement, so the amplitude is arbitrary within classical mechanics. In quantum mechanics, however, the vibrations should be quantized into phonons. This can be performed by a standard procedure where the motion of the atoms is resolved into normal modes. Each mode resembles a harmonic oscillator, which we shall solve in Section 4.3. Unfortunately this procedure is beyond the scope of this book, and we shall take a shortcut.

The amplitude U_0 can be found for a single phonon by calculating the total energy of the vibration and equating it to the quantized value $\hbar\omega_q$. For the monatomic crystal, the velocity of atom j is given by

$$\frac{du_j}{dt} = U_0\omega_q \sin(qaj - \omega_q t), \quad (2.28)$$

so the kinetic energy is $\frac{1}{2}mU_0^2\omega_q^2 \sin^2(qaj - \omega_q t)$, whose average value is $\frac{1}{4}mU_0^2\omega_q^2$. This equals the average potential energy for harmonic motion, so the total average energy is $\frac{1}{2}mU_0^2\omega_q^2$ per atom. Multiplying by the number of atoms N_{atoms} yields the total energy, and equating this to the quantum $\hbar\omega_q$ fixes U_0 . We can eliminate N_{atoms} in favour of the 'volume' Ω (really a length), the mass density ρ , and the mass of each atom m , as $N_{\text{atoms}} = \Omega\rho/m$. Thus

$$U_0 = \sqrt{\frac{2\hbar}{N_{\text{atoms}}m\omega_q}} = \sqrt{\frac{2\hbar}{\Omega\rho\omega_q}}. \quad (2.29)$$

For small wave vectors, where $\omega_q = v_s q$, this becomes $U_0 = \sqrt{2\hbar/\Omega\rho v_s q}$. The motion of an atom is finally

$$u_j(t) = \sqrt{\frac{2\hbar}{\Omega\rho\omega_q}} \cos(qaj - \omega_q t). \quad (2.30)$$

Note the appearance of the volume Ω of the crystal. A larger sample appears to have 'weaker' phonons but this is cancelled by the increase in the total density of states. We shall see that Ω vanishes from physical results such as scattering rates.

2.8.3 DIATOMIC CHAIN

Many crystals, including the common semiconductors, have more than one atom per unit cell, which introduces another feature. Consider the chain shown in Figure 2.23(b). This has alternating heavy and light masses, M and m , along its length.

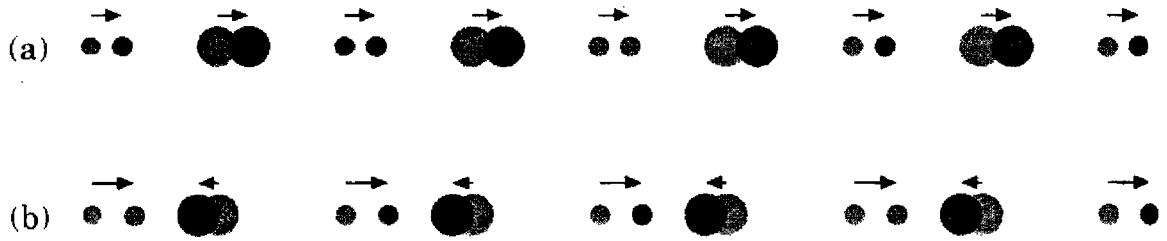


FIGURE 2.24. Motion of the atoms in (a) acoustic and (b) optic modes of very long wavelength for a one-dimensional chain of alternating light and heavy atoms. The light grey discs mark the resting positions of the atoms.

giving two atoms per unit cell. The lattice and spring constants retain their previous values a and K . The dispersion relation is

$$\omega_q^2 = K \left(\frac{1}{m} + \frac{1}{M} \right) \pm K \sqrt{\left(\frac{1}{m} + \frac{1}{M} \right)^2 - \frac{4}{mM} \sin^2 \frac{qa}{2}}. \quad (2.31)$$

There are now two bands or branches to the dispersion relation. The lower or acoustic branch looks much like the monatomic case, except that the increased mass has lowered the frequencies. However, the upper or optic branch is quite different. It has a maximum at $q = 0$ and slopes down gradually to the zone boundary, where there is a gap between the acoustic and optic branches. The amplitudes of vibration of the two atoms in each unit cell of the diatomic crystal can be calculated as for the monatomic chain.

The natures of the vibrations near $q = 0$ in the two branches are contrasted in Figure 2.24. The displacements in the acoustic branch are like sound waves, as the name implies. The two atoms in each unit cell move in the same direction by almost the same distance, and over a small region it appears as if the whole crystal has simply been compressed or stretched. We shall use this picture when calculating the interaction between such phonons and electrons in Section 8.4.1. The distortion of the bonds between the atoms becomes smaller for a given amplitude of vibration as $q \rightarrow 0$, sending $\omega_q \rightarrow 0$.

The heavy and light atoms move in opposite directions in the optic case, as in Figure 2.24(b). This is very similar to the top of the band for the monatomic chain. Again there is maximum distortion of every spring, so the frequency takes its maximum value.

A further feature appears in compounds because of the transfer of charge between the two species. Relative displacement of the two atoms, as in the optic mode, sets up an electric dipole $\mathbf{p}$. This in turn sets up a polarization field $\mathbf{P}$ and an electric field which can interact with electromagnetic waves, hence the name 'optic'. Charged particles such as electrons also feel the electric field, and we shall see in Section 8.4.2 that polar longitudinal optic (LO) phonons scatter electrons rapidly. This polar nature is not seen in elements because all the atoms are identical and there is no transfer of charge.

Only one type of atom moves in each of the modes at the zone boundary, the other remaining stationary. For example, only the lighter atoms vibrate in the upper mode, with adjacent cells in antiphase. Thus only m appears in ω_q , and the expression is the same as that for the highest frequency in the monatomic chain, except that the effective value of K is halved because there are now two springs separating each light atom. This behaviour is analogous to that for electrons at zone boundaries, shown in Figure 2.4.

2.8.4 PHONONS IN THREE DIMENSIONS

The band structure for phonons can be plotted in exactly the same way as that for electrons. A small part of the dispersion is sketched in Figure 2.25 for Si and GaAs. The semiconductors have two atoms in each primitive unit cell, so the overall structure is close to that for the diatomic chain in Figure 2.23(b). The frequencies are higher for Si largely because of its lighter mass, which is 28, compared with an average of 72 for GaAs. The main difference from the one-dimensional case arises from the three possible polarizations. The two transverse modes are degenerate along [100] but in general there are three acoustic and three optic branches. Both acoustic modes become linear for small q , the longitudinal mode having a higher velocity.

The transverse optic (TO) and LO phonons in polar semiconductors have slightly different energies at Γ . This splitting is due to the electric field set up by LO phonons and is given by the Lyddane–Sachs–Teller relation (8.60). The energy of the LO phonon is 36 meV in GaAs; this high energy means that LO phonons can be

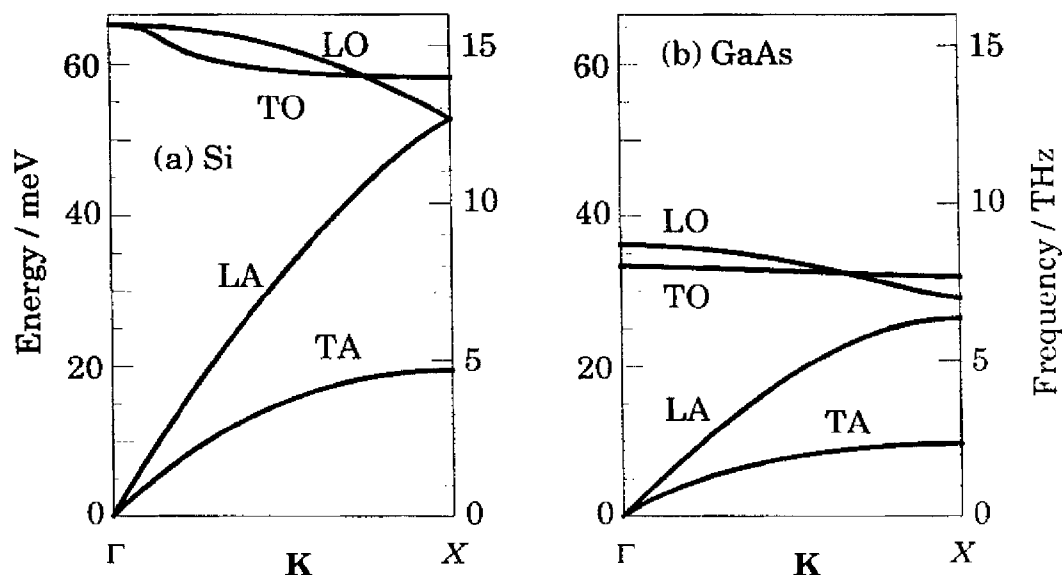


FIGURE 2.25. Sketch of the dispersion relation for phonons in (a) Si and (b) GaAs from Γ to X . The transverse branches are doubly degenerate along this direction.

neglected at low temperatures (below about 77 K), but they are an important mechanism for transferring energy between electrons and phonons at higher temperatures. The lighter mass of AIAs raises this energy to about 50 meV, but the energy of LO phonons in $\text{Al}_x\text{Ga}_{1-x}\text{As}$ does not simply change smoothly from one to the other as a function of x .

This concludes our survey of the crystalline semiconductors. In the next chapter we shall see how these ingredients are put together to make heterostructures and investigate the properties that result.

Further Reading

Band structure is covered in all textbooks on solid state physics, including the classics by Kittel (1995) and by Ashcroft and Mermin (1976). They tend, however, to concentrate primarily on metals rather than semiconductors. A newer comprehensive introduction is by Myers (1990), and Bube (1992) is more biased towards semiconductors.

Seeger (1991) gives a broad review of classical semiconductor physics including the effect of non-parabolic valence bands. Wolfe, Holonyak, and Stillman (1989) present a clear account of the physical properties of semiconductors. A thorough description of band structure and group theory, heavily biased towards semiconductors, is given by Yu and Cardona (1996). They also cover phonons and electronic and optical properties with a chapter on heterostructures.

A wealth of data on semiconductors is tabulated by Madelung (1996).

EXERCISES

- 2.1** Show that the two statements of Bloch's theorem, equations (2.2) and (2.3), are equivalent.
- 2.2** Figure 2.5 was drawn for a typical band in a semiconductor. Repeat this for a superlattice, which might have $a = 10 \text{ nm}$ and $W = 50 \text{ meV}$. What values of effective mass and maximum velocity are expected?
- 2.3** Many semiconductors have a saturation velocity for electrons of around 10^5 m s^{-1} . How much of the Brillouin zone do they need to explore to reach this velocity? The saturated velocity is reached at a field of around 10^6 V m^{-1} , so roughly what is the relaxation time?
- 2.4** Calculate the frequency of Bloch oscillation in as large an electric field as the material will tolerate before breaking down, and the corresponding excursion in space. Use the parameters of Figure 2.5. Compare this with a typical lifetime between scattering events (estimated from the mobility



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